

FINAL PROJECT DOCUMENTATION

Electric Vehicle Analytics Dashboard using Flask & Tableau

E-CarStart

Academic Project Report

Abstract

This project presents an Electric Vehicle Analytics Dashboard developed using Python Flask and Tableau Public. The application provides interactive visualizations to analyze EV registrations, manufacturer performance, vehicle categories, and adoption trends through a responsive web interface.

Introduction

The rapid growth of electric vehicles has increased the need for accessible data analysis tools. This project integrates Flask and Tableau to present EV insights in an easy-to-use web application.

Problem Statement

EV data is often dispersed across multiple sources, making it difficult for users to gain meaningful insights. The project addresses this by providing a centralized analytics platform.

Objectives

- Develop an interactive EV analytics website.
- Integrate Tableau dashboards and stories.
- Provide responsive navigation.
- Enable easy exploration of EV trends.

Existing System

Existing EV information is available through reports and isolated portals with limited interactive capabilities.

Proposed System

The proposed E-CarStart platform embeds Tableau dashboards within a Flask application, enabling users to interactively explore EV data.

Technologies Used

Frontend: HTML, CSS, Bootstrap

Backend: Python Flask

Visualization: Tableau Public

IDE: VS Code

Version Control: Git & GitHub

System Architecture

Presentation Layer (HTML/CSS/Bootstrap) → Flask Application Layer → Tableau Public Visualizations → EV Dataset.

Implementation

Flask routes manage navigation between Home, Dashboard, Story, and About pages.

Tableau Public dashboards are embedded using the Tableau JavaScript API.

Testing and Results

Functional, performance, browser compatibility, and responsive testing were conducted.

All modules operated successfully and provided expected outputs.

Advantages

Interactive visualization, responsive interface, modular design, easy maintenance, improved accessibility to EV insights.

Future Scope

Add live datasets, predictive analytics, user authentication, downloadable reports, and cloud deployment.

Conclusion

The Electric Vehicle Analytics Dashboard successfully demonstrates how Flask and Tableau can be combined to deliver a modern analytics platform for exploring India's EV ecosystem.

References

1. Flask Documentation
2. Tableau Public Documentation

3. Bootstrap Documentation

4. Python Documentation